



Amazon Reviews

Visualization Course

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Preface

Amazon Product Reviews Project (1996-2023)

The [Amazon Reviews](#) project, collected by [McAuley Lab](#), presents a comprehensive and large-scale dataset of customer reviews collected from Amazon, covering a wide range of products over several decades. It features rich metadata such as item descriptions, prices, images, review timestamps, and user feedback (ratings). The project provides clear documentation, data access instructions, and ready-to-use formats (such as json, or jsonl), making it easier for researchers and developers to explore and analyze the dataset.

Research Methods

Our research will be based on **Amazon Reviews 2023** dataset, collected in **2023** by [McAuley Lab](#), at UC San Diego by PI Prof. Julian McAuley.

In this work we focused on Pure IDs (5-Core), subset of the data in which all users and items have at least 5 reviews this gives higher quality reviews, reduced noise, balanced distribution and computational efficiency.

https://amazon-reviews-2023.github.io/data_processing/5core.html

The data we used is product based (items) and user meta-data. It's suitable to explore user behavior and understand how user preferences change over time.

The data composed of:

User Reviews (user ratings, titles, dates, etc.)

- user id
- product code (parent asin)
- ratings
- timestamp

Item Metadata (price, images, descriptions, etc.)

- title (name of the product)
- price \$ (at time of crawl)
- average rating
- rating number
- descriptions
- price
- store
- details etc.

Problem Statement Story

Understanding user ratings is critical for e-commerce platforms and artists as they reflect consumer sentiment, satisfaction, and product success. However, rating distributions can be skewed or influenced by various factors, such as time, price, and total rating count, making it challenging to extract actionable insights or detect patterns over years. In the context of Amazon's Digital Music Category, we aim to systematically analyze and visualize users' ratings and cd titles to reveal underlying trends and relations between the columns.

Hypothesis and Research questions

Research Questions

- How are user ratings distributed in the Digital Music category on Amazon?
- How have average user ratings changed over years, and what trends can be observed?
- Is there a relationship between the price of a product and the ratings it receives?
- What patterns emerge when comparing individual ratings to the product's average rating and total number of ratings?
- Do we observe biases, such as a predominance of high ratings, or period-specific changes?

Hypothesis:

H1: Most ratings will cluster around 4 and 5 stars, indicating a positive bias or general satisfaction with digital music purchases.

H2: There is a temporal trend in average ratings such as a decline or increase over certain periods which may reflect changes in consumer expectations, the evolution of Amazon's platform, or shifts in the types of products offered.

H3: Higher-priced digital music products will receive higher average ratings, suggesting that users perceive expensive albums as higher quality.

H4: Special events such as sales, holidays, and public celebrations significantly influence the price distribution of top-selling CDs, potentially creating distinct pricing patterns across different event types.

Data Exploration

Dataset Overview

Core-5 Version

The “Amazon Review 2023” project also provides a “**5-Core**” dataset version, which includes only users and items that each have at least 5 reviews. It contains higher quality reviews, reduced noise, balanced distribution and computational efficient for small-mid size projects. In this version:

- **User threshold** – each reviewer contributes ≥ 5 reviews.
- **Item threshold** – each product accrues ≥ 5 reviews.

Benefit	Why it matters
Higher-quality signals	Low-activity users/items—often noisy—are removed.
Better balance	Ratings are more evenly distributed across users and products.
Leaner size	Fits comfortably into small-to-mid-scale projects without heavy infrastructure.

Category Overview

Category	#Users	#Items	#Ratings
CDs_and_Vinyl	123.9K	89.4K	1.6M

The dataset contains many categories; however, we have decided to focus on the “**CDs_and_Vinyl**” category for our project. It has 123k user ratings over 89.4K items with total of 1.6M ratings. (each user can rate multiple items).

Data Source

- “CDs and Vinyl” **5-Core** dataset can be downloaded [here](#).
- “CDs and Vinyl” **Meta** dataset can be downloaded [here](#).

Note: the meta dataset is complementary dataset (and the biggest one). To merge them both, we use the “parent_asin” field to join between the two datasets.

Dataset Description

Data Types

Our **non-trivial** “CDs and Vinyl” 5-Core dataset contains the following data types:

#	Column	Non-Null Count	Dtype
0	user_id	1552764 non-null	object
1	parent_asin	1552764 non-null	object
2	rating	1552764 non-null	float64
3	timestamp	1552764 non-null	int64
4	main_category	1552090 non-null	object
5	title	1552764 non-null	object
6	average_rating	1552764 non-null	float64
7	rating_number	1552764 non-null	int64
8	features	1552764 non-null	object
9	description	1552764 non-null	object
10	price	1451308 non-null	object
11	images	1552764 non-null	object
12	videos	1552764 non-null	object
13	store	1552707 non-null	object
14	categories	1552764 non-null	object
15	details	1552764 non-null	object
16	bought_together	0 non-null	float64
17	subtitle	11802 non-null	object
18	author	11160 non-null	object

dtypes: float64(3), int64(2), object(14)

Dataset shape:

Rows: 1,552,764

Columns: 19

The dataset contains several times such as Objects, Float64, Int64, Strings.

Note: pandas dataframe will generalize a feature as object while it's being a number or a string or array, (e.g string => object). This will be fixed (as described in the data cleaning section).

Data Formats:

First, we take a look what a sample row looks like:

user_id	AFWHJ6O3PV4JC7PVOJH6CPULO2KQ
parent_asin	B000002AGY
rating	5.0
timestamp	1358286606000
main_category	Digital Music
title	Concierto
average_rating	4.7
rating_number	396
features	[]
description	[]
price	4.0
images	[{'thumb': 'https://m.media-amazon.com/images/...'}]
videos	[]
store	Jim Hall Format: Audio CD
categories	[CDs & Vinyl, Today's Deals in Music, Jazz, Ja...]
details	{}
bought_together	NaN
subtitle	NaN
author	NaN

We can already learn that:

1. Some features aren't formatted correctly (e.g. "price" it's marked as object instead of float64) (or contains invalid characters, e.g. '\$')
2. Some features contain a list of items (for example images).
3. Some features can be dropped as they are irrelevant (e.g. Videos)

Notes:

1. We will convert the types (as pandas dataframe doesn't infer types correctly)
2. We will create/add new features (like datetime instead of timestamp)
3. We will calculate new features (for more analysis)

Cleaning The Data

Identifying the Category

After understanding how the data behaves in general, we'll begin analyzing and cleaning it a bit. First thing we do it count how many items we have per main category inside the **CDs and Vinyl** dataset:

main_category	
Digital Music	1508950
Movies & TV	30433
Books	12377
Toys & Games	99
Office Products	58
Cell Phones & Accessories	43
Amazon Home	27
All Beauty	25
Sports & Outdoors	16
Industrial & Scientific	14
Tools & Home Improvement	13
AMAZON FASHION	11
Video Games	9
Home Audio & Theater	9
Musical Instruments	6

dtype: int64

From this table we can see that the main category is “**Digital Music**”, where the rest are either not relevant or barely adds any kind of extra information to the rest of our analysis. So, we will drop all the items which are not “Digital Music”.

Invalid data

Next, we will look for invalid data (such as NaN, Null). Removing these values is usually safe and beneficial, as it improves data integrity, ensures statistical accuracy and contributes to a clearer understanding of the information.

	Missing Count	Missing %
bought_together	1552764	100.00
author	1541604	99.28
subtitle	1540962	99.24
price	101456	6.53
main_category	674	0.04
store	57	0.00

The missing-value snapshot reveals that three columns: **bought_together**, **author**, and **subtitle** are almost completely empty (100 %, 99.28 %, and 99.24 % missing), so they add no analytical value unless we can enrich them from another source and are therefore good candidates for outright removal.

In contrast, **price** is absent in only about 6.5 % of the rows; rather than discarding a useful feature, we can either impute the gaps (for example, with the median price within each *main_category*) or keep a binary “price_missing” flag for modelling.

The remaining two fields, **main_category** and **store**, are nearly intact ,missing in just 0.04 % and virtually 0 % of records so it is easier and safer to either drop those few incomplete rows or fill them with a neutral label such as “unknown.” Cleaning the data along these lines will help us improve integrity, reduce noise, and give us a leaner, more reliable dataset for downstream analysis or modelling.

Cleaning Steps

1. **Filtering:** Use only the “Digital Music” category. As it contains most of the data. this leaves us with 1508950 rows and 19 columns.
2. **Cleaning Titles:** Some titles contain extra spaces, weird names (e.g. “OSCD AD”), some contains bad formatting (e.g. The Four Horsemen[explicit]), some contains long CD titles. Invalid CD Titles (e.g. 4:44, 50, 405), some contains empty or 1 character (e.g. “), some contains dates (e.g. 27-Jul). These types of titles are not valid as they do not contain valid names.
3. **Remove rows:** remove invalid data (e.g. Title, Price).
4. **Converting:**
 - a. **Rating:** convert to numeric
 - b. **Average_Rating:** convert to numeric
 - c. **Price:** convert to numeric
 - d. **Rating_Number:** convert to to Numeric
5. **Creating new Columns:**

Compute columns that will be used for future timeseries analysis.

 - a. **Star:** values between [1,2,3,4,5] based on rating column rounding the rating values to flat rating.
 - b. **Datetime:** Date object, converted from “Timestamp” column.
Note: Timestamp is in Unix format.
 - c. **Season:** Winter, Spring, Summer, Fall
 - d. **Date:** A Date format instead of Unix Timestamp.
 - e. **Weekday:** The number of the weekday
 - f. **Hour:** Hour by the date.
 - g. **Month:** Month by the date.
 - h. **Year:** Year by the date.
 - i. **Special Event:** “Black Friday”, “Christmas Eve”, ...etc.
6. **Rename Columns:**
 - a. **Rating_Number** was renamed to **Total_Ratings**.
It’s easier to call it total rating than “Rating Number”.
7. **Drop Columns:**

Dropping the columns get rid of unused features and reduce clutter and noise.

 - a. timestamp
 - b. images
 - c. features
 - d. videos
 - e. subtitle
 - f. bought_together
 - g. author
 - h. description
 - i. details

After the cleaning, we remain with 12 columns and 1,508,950 rows.

The updated dataset now looks like this:

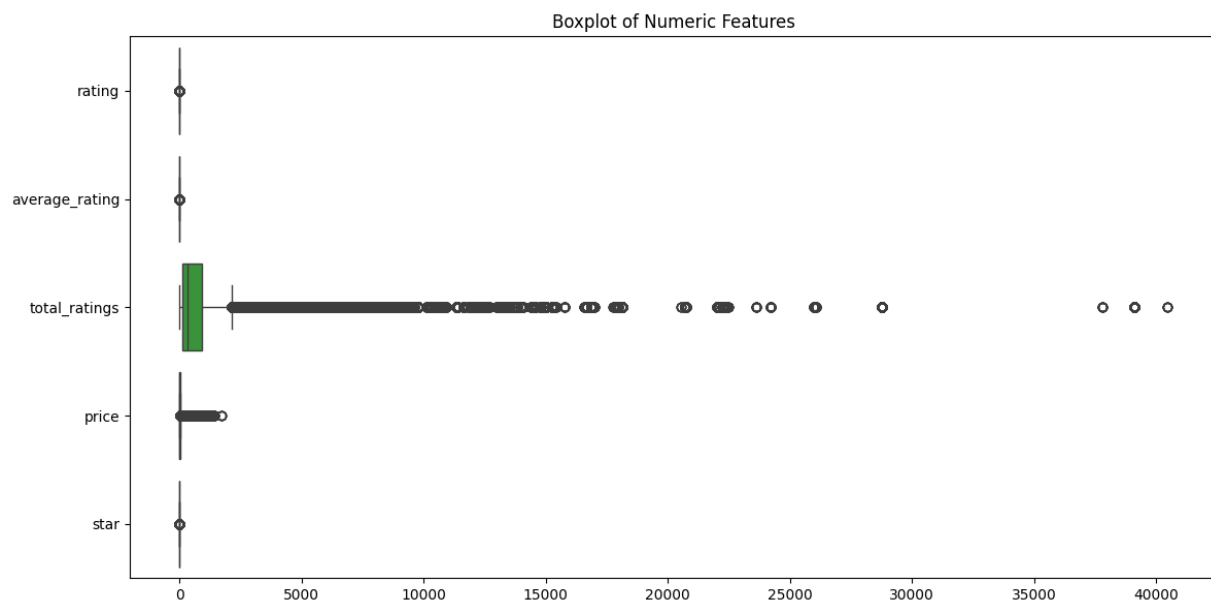
Data columns (total 18 columns):				
#	Column	Non-Null Count		Dtype
0	user_id	1508950	non-null	object
1	parent_asin	1508950	non-null	object
2	rating	1508950	non-null	float64
3	main_category	1508950	non-null	object
4	title	1508950	non-null	object
5	average_rating	1508950	non-null	float64
6	total_ratings	1508950	non-null	int64
7	price	1416492	non-null	float64
8	store	1508950	non-null	object
9	categories	1508950	non-null	object
10	star	1508950	non-null	int64
11	datetime	1508950	non-null	datetime64[ns]
12	weekday	1508950	non-null	object
13	hour	1508950	non-null	int32
14	month	1508950	non-null	int32
15	year	1508950	non-null	int32
16	season	1508950	non-null	object
17	special_event	643909	non-null	object

Note: Pandas shows a column as having the object data type when it contains text or a mix of different kinds of data. But sometimes, even columns that are supposed to be numbers or dates end up as object if they contain things like symbols, inconsistent formatting, or missing values written as text like "N/A" or "null".

Exploratory Data Analysis (EDA)

In this section, we will analyze and present the findings regarding the features (columns) that our dataset contains. Exploratory Data Analysis (EDA) is a critical step in any data analysis or machine learning workflow. It allows us to better understand the structure, patterns, and relationships within our data before applying any models or making decisions

Outliers Visualization:



We can learn from this box-plot graph that:

total_ratings - has a wide range with many extreme outliers. The IQR (interquartile range) is small compared to the spread of data, most values are tightly packed near the bottom. This indicates that most items have few ratings, but a few items are extremely popular with tens of thousands of ratings (long tail distribution).

price - The price column also has many outliers on the high end. Most products are priced low (as the box is very compressed), but there are a few very expensive items. This suggests right-skewed pricing, typical in e-commerce datasets.

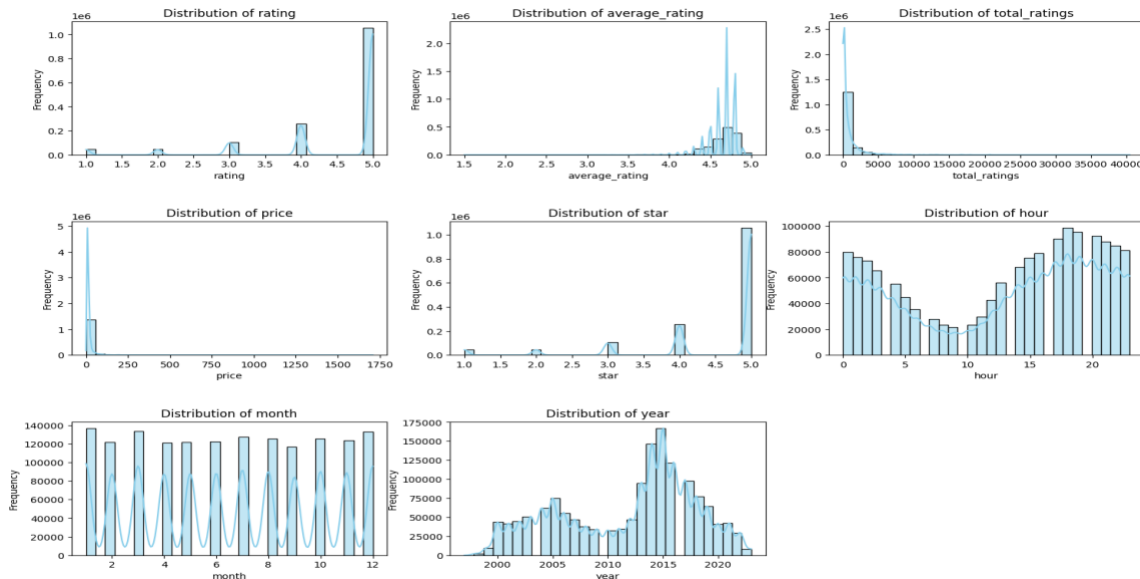
rating, average_rating, star - Are Well-Distributed.

These columns have relatively tight boxes and fewer outliers. It implies that most ratings are clustered within a small range probably between 3 and 5. For star, being integer-based (1–5), the distribution appears compact and normal.

Price column- at first, we tried to use forward fill to compute each product, but we discovered that there are products that did not have any price from first day, there are 30,247 CDs, therefore we decided to drop null column and not to use other computation methods.

Data Distribution

These plots allow us to view the distribution of the data visually. It helps us to understand how each feature behaves. (e.g. a normal behaviour always follows a gaussian distribution)



Understanding the numerical columns: (mean, median, min, max, std):

	count	mean	std	min	25%	50%	75%	max
rating	1416492.00	4.48	0.97	1.00	4.00	5.00	5.00	5.00
average_rating	1416492.00	4.65	0.17	1.50	4.60	4.70	4.80	5.00
total_ratings	1416492.00	931.33	1909.99	4.00	104.00	323.00	943.00	40466.00
price	1416492.00	18.34	30.33	0.01	8.99	12.53	17.98	1708.76
star	1416492.00	4.48	0.97	1.00	4.00	5.00	5.00	5.00
hour	1416492.00	12.83	7.50	0.00	5.00	15.00	19.00	23.00
month	1416492.00	6.47	3.49	1.00	3.00	6.00	10.00	12.00
year	1416492.00	2011.81	6.15	1997.00	2006.00	2014.00	2016.00	2023.00

What can we quickly learn from the data?

1. The average price of CDs (or Vinyl) rated by users are 18.34\$.
2. Some titles with price of 0.01\$ are meant for clearing (or buying them is 1cent but shipping will cost 3-5\$).
3. Most of the rated CDs and Vinyl's were done in 2015 (most active users).
4. The best months were December (Christmas) and January (after New-Year).
5. Most rated CDs and Vinyl's were given 5 Stars.
6. The average rating is between 4.5 and 5.0.

Feature Correlations

At this stage, we will create a correlation matrix to analyze the relationships between all numerical features in the dataset. This analysis is important because it helps us identify which variables are strongly related either positively or negatively, which can reveal underlying patterns in the data. Understanding these correlations can guide us in feature selection, detect multicollinearity, and help shape the direction of further analysis or model building.

For example, it may help us reduce the number of dimensions, or see which features interact best with which features.



- Moderate correlation between average_rating and total_ratings (0.44), Popular items (more total ratings) tend to have **higher average ratings**.
- Negative correlation between total_ratings and price (-0.19), Cheaper products tend to get more reviews.
- Slight correlation between year and features like rating, total_rating, star, suggests some temporal trends (e.g., newer products may be rated differently or attract more engagement).

Hypothesis Exploration

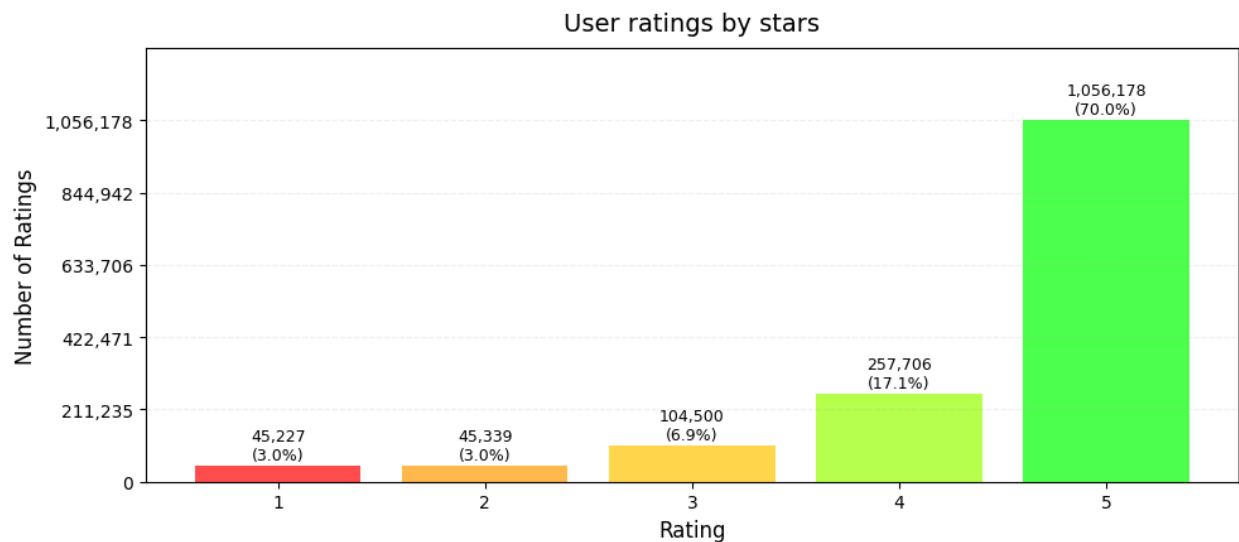
Forming a **hypothesis** is important because it gives the analysis a clear direction and purpose. Rather than just creating random charts, a hypothesis helps focus on specific questions or assumptions you want to test using the data. It guides what to look for, which variables to explore, and how to interpret the visual results

Hypothesis 1- Rating Distribution by Stars

The first plot shows the ratings per star, the idea is to explore the distribution of ratings and according to the chart it is highly skewed toward the positive end, most users award 4 or 5 stars rating. About 3% of people gave low rating of 1 or 2 stars, 7% were neutral with rating of 3, while 17% gave rating of 4, the majority 70% gave 5 rating.

Nearly 90% of digital music user ratings are 4 or 5 stars, confirming a strong positivity bias among purchasers. Negative ratings are rare, highlighting either a generally high satisfaction or a tendency for only satisfied users to leave reviews.

Plot 1



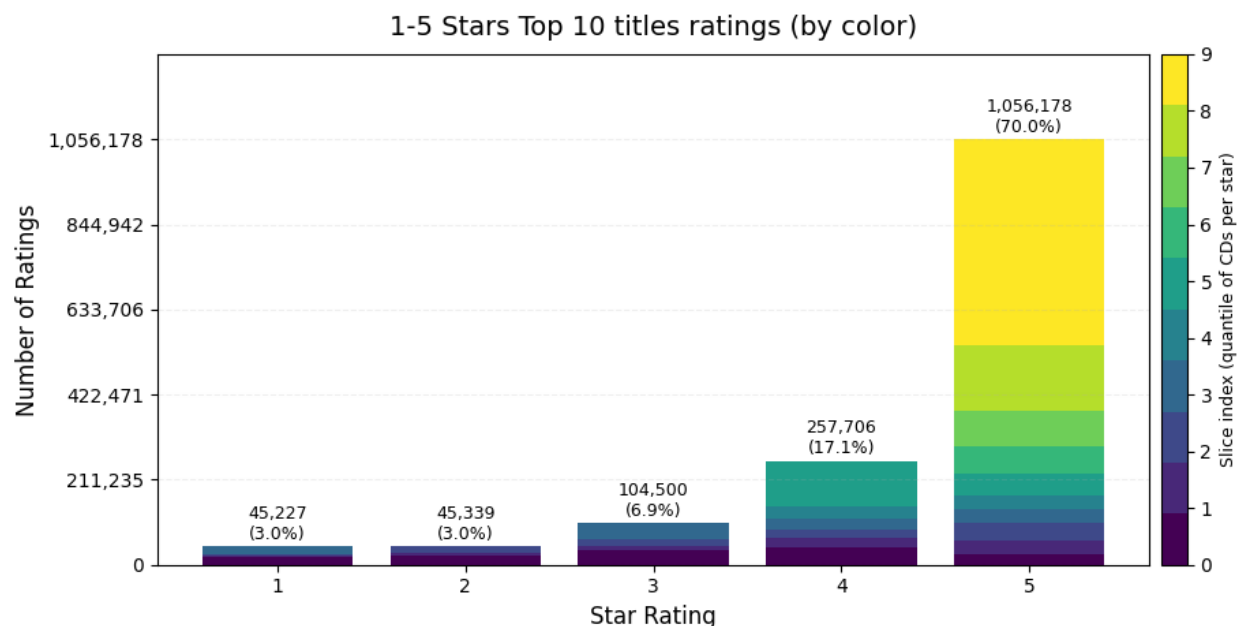
The clustering around the top star ratings suggests either:

- A positive customer experiences
- Leniency or positive bias in user reviews for digital music.
- Or possibly that dissatisfied users rarely leave ratings.

The second plot is a “Slice Visualization” that shows the top 10 CDs as slices in each star category (not all the star categories contain top 10), It adds granularity by revealing how the star ratings are distributed among the top CD’s. This deepens our understanding of the digital music rating landscape. After observing in Plot 1 that nearly 90% of all ratings are highly positive, we dissected which products drive those ratings in Plot 2.

The stacked bars show that for each star category, a small number of CDs capture most of the ratings, especially at the 5-star level. This highlights not only a strong positivity bias, but also a clear concentration of popularity the streaming landscape is shaped by a few blockbusters, while most titles see only modest engagement, even among top-rated products. Consequently, both consumer satisfaction and market activity are ‘top-heavy’ positive reviews abound, but attention and acclaim are focused on a handful of leading albums

Plot 2



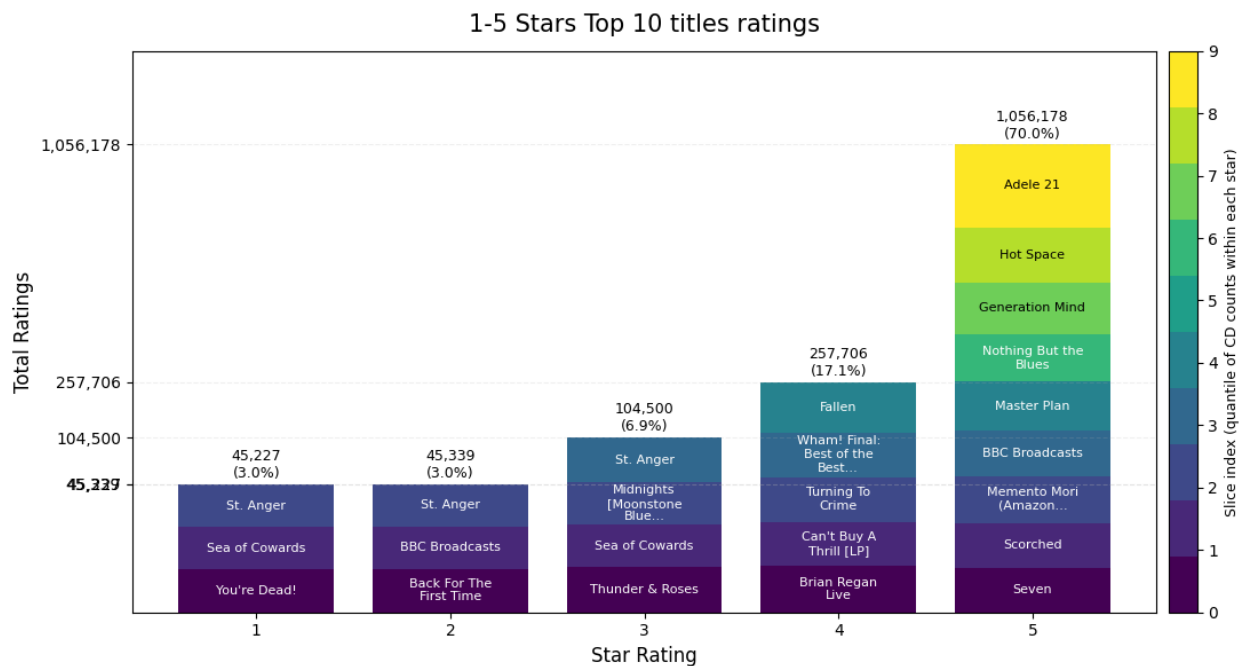
We observe that the 1-3 Star category are rare according to the bars that are much shorter, confirming the strong positive skew seen in Plot 1.

In the 5-star bar, the color gradient suggests that even among highly rated CDs, a few titles receive a disproportionately large number of ratings (the larger, higher “slice index” segments).

While we see distribution patterns and relative “weight” of ratings by slice/quantile, we cannot tell which exact CD corresponds to each colored slice/bar segment. This makes it harder to link a segment directly to a particular title without supplemental annotation.

The third plot, allows to deepen the picture, showing exactly which titles dominate the star system and how concentrated user enthusiasm is. A small handful of releases do not just collect the most ratings; they collect almost all the highest ratings. At the same time, negative ratings (1, 2 star) are rare and fragmented, reinforcing the market's bias toward rewarding popular titles and shielding them from criticism.

Plot 3



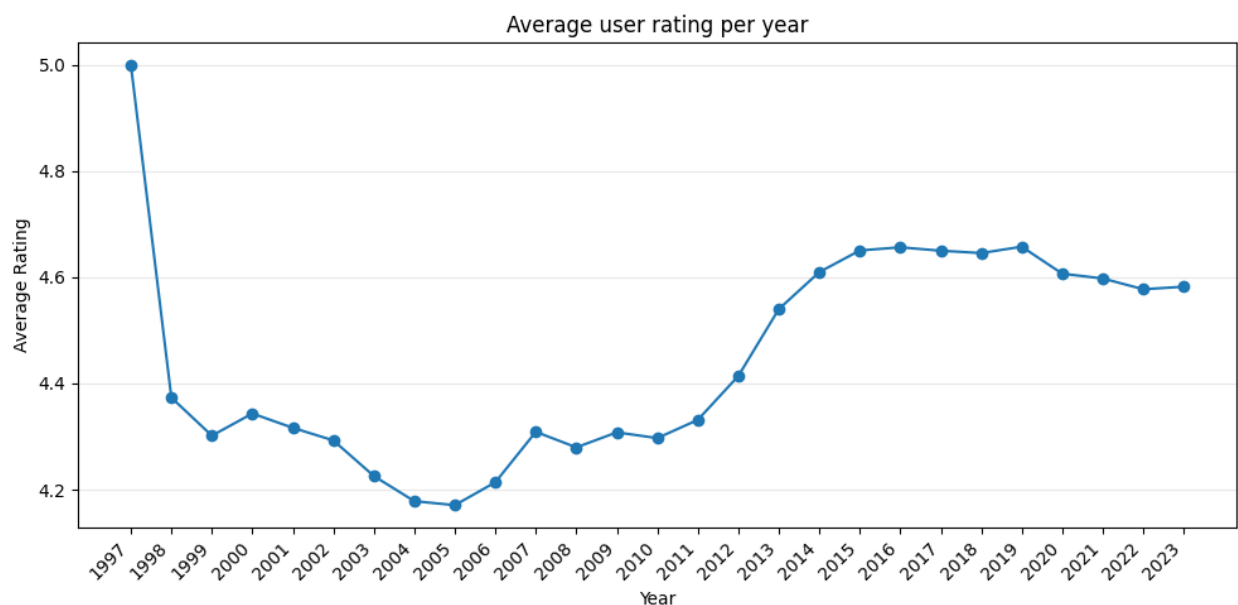
Through these three visualizations, a clear narrative emerges: Digital Music ratings are both extremely positive and highly concentrated. Most users give 4 or 5 stars, but it's a small set of superstar CDs that attract the lion's share of these positive reviews. Negative ratings are rare and scattered across less-visible releases.

As a result, standout artists not only dominate the charts, but they also shape the perception of digital music as a field defined by satisfaction and blockbuster appeal. Each subsequent plot has peeled back another layer, moving from overall sentiment (Plot 1) to concentration among top releases (Plot 2), to a detailed breakdown showing just how dominant these few titles are within each star rating (Plot 3).

Hypothesis 2 - Time Analysis trends by rating and reviews

We wanted to measure the rating over time, to see if we can find correlations over time. The line chart illustrating average user ratings per year from 1997 through 2023 reveals notable temporal trends in customer sentiment. In the earliest year of available data, 1997, the average rating reaches the maximum of 5.0; however, this likely reflects a very low volume of early reviews, often from perfect scorers. Following this, average ratings exhibit a decline starting in 1998, continuing through the early 2000s and reaching a minimum around 2005–2006 of approximately 4.18. After this low point, ratings gradually recover, increasing steadily to peak between 2016 and 2019 at around 4.65. Since then, the average rating has experienced a mild plateau or slight decline, settling at about 4.59 in 2023. Throughout the past decade, from 2013 to 2023, ratings have remained solidly above 4.5, indicating both long-term stability and resilience in overall positive user sentiment.

Plot 4



Average digital music user ratings declined following Amazon's launch years, reached a nadir during the mid-2000s, and have steadily rebounded to high, stable levels since the 2010s. This U-shaped (decline, then rise, then plateau) trend reflects evolving user expectations and platform development, supporting the hypothesis that rating patterns shift in parallel with product and platform evolution.

The next plot visually extends the examination of user satisfaction over time and deepens the evolving narrative of the rating ecosystem:

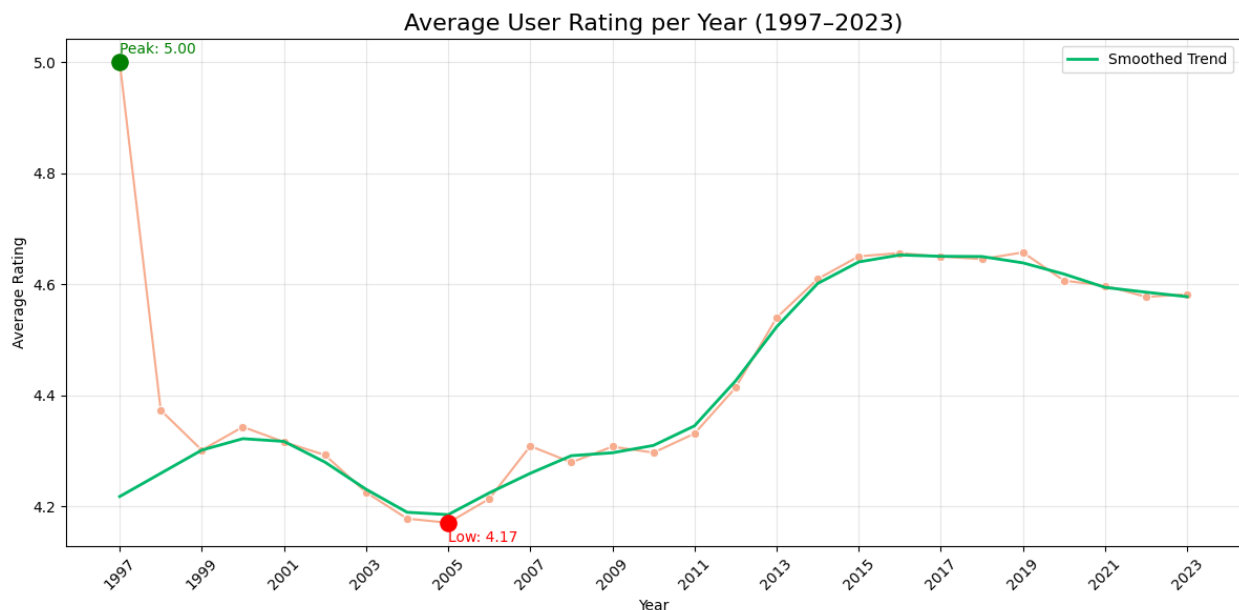
Early Phase (1997–2000): Sparse data, early adopters, or less critical reviewers could explain high ratings.

Decline (2001–2006): As more users joined and selection broadened, standards or expectations may have increased, leading to more critical ratings, or perhaps Amazon's review tools filtered less stringently.

Recovery and Growth (2007–2019): Improvements to the platform (review policies, rating guidance, overall customer experience) and the refinement of curation/offerings may have reversed the trend, returning satisfaction to higher levels.

Recent Plateau: The flattening since 2019 might reflect a mature platform, stable user base, and less room for significant change.

Plot 5



This plot delivers a narrative arc: the digital music rating ecosystem, initially buoyed by a wave of perfect scores, undergoes a period of critical adjustment as it expands, then recovers as both consumers and the platform evolve together. Where the earlier plots established the sheer dominance of positive sentiment and identified the titles or slices responsible, this temporal view demonstrates that such positivity is not static or inevitable. Instead, it is the result of a dynamic interplay of consumer expectations, platform evolution, and market development first challenged during the platform's rapid scaling, then reaffirmed as a stable, collective experience in the more recent decade.

Adding the review dimension as well we discovered that the extremely high average rating (close to 5.0 in the earliest year), which rapidly drops in the first few years. This early spike is likely due to a very small sample of enthusiastic or "perfect" reviewers when Amazon's music review system was first introduced.

Decline and Bottoming Out (Early Mid 2000s):

Average ratings decrease steadily and reach their lowest point around the mid-2000s (about 4.18), coinciding with growing platform usage and possibly more critical or diverse reviewer participation.

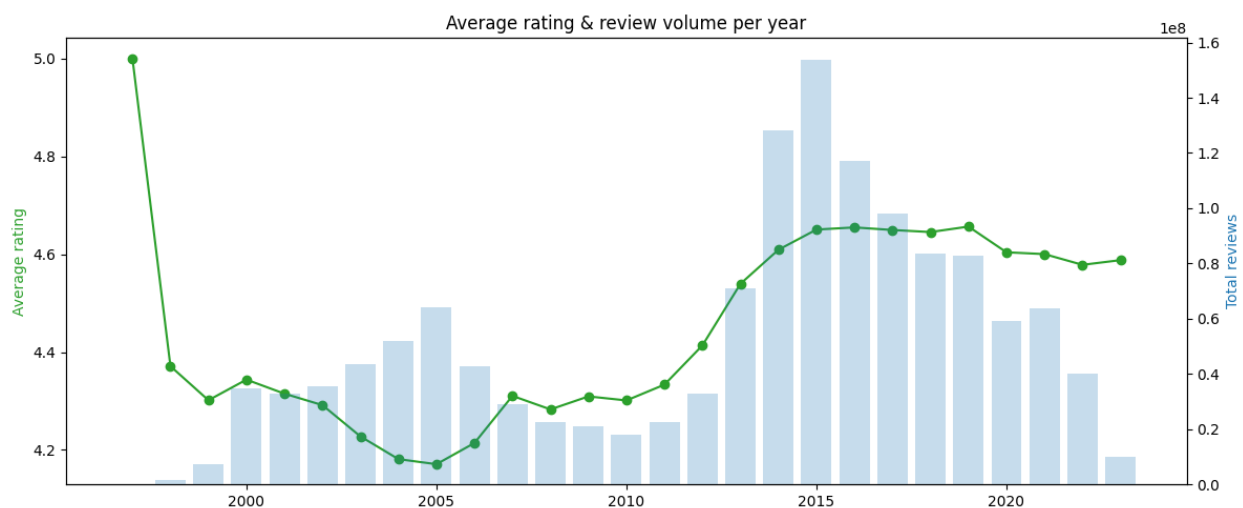
Gradual Recovery and Plateau (2010s):

From the late 2000s onward, the average rating starts to rise again, reaching a relatively stable, high plateau above 4.5 starting around 2013 through to the early 2020s. This corresponds to the peak in review volume, indicating a mature and active marketplace with a well-established user base.

Slight Decline or Stabilization (2020s):

In the most recent years, both average ratings and the number of reviews show a plateau or slight decline. This may reflect market saturation, changing consumer behaviours, or external factors (such as broader platform or market shifts).

Plot 6



Finally, Positive Bias the long-term trend reaffirms that digital music ratings on Amazon are highly positive, most often clustering above 4.5 consistent with the first hypothesis about ratings bias and high user satisfaction.

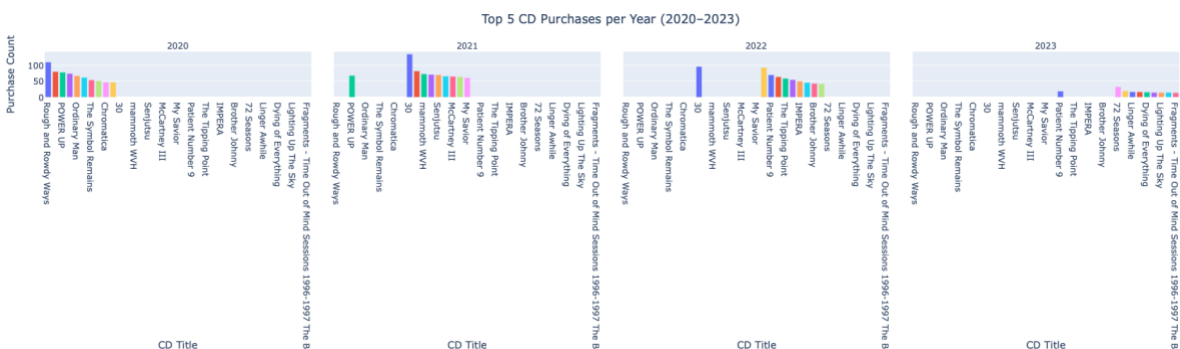
Where earlier analyses focused on the distribution and concentration of positive ratings, this chart adds a crucial temporal dimension, demonstrating that high user satisfaction is not merely a static feature, but the outcome of platform growth, community adaptation, and evolving customer expectations.

Hypothesis 3 - Time Analysis trends by purchase and price

After establishing that digital music ratings are overwhelmingly positive and that a small group of albums consistently dominates both ratings and purchase counts, we explored whether pricing plays a role in user perception and satisfaction. The faceted bar chart below (Plot 7) highlights the top 5 CD purchases for each year from 2020 to 2023. Across these years, several premium-priced albums appear consistently at the top, suggesting not only sustained popularity but also that consumers are willing to invest in what they perceive as higher-quality digital music experiences.

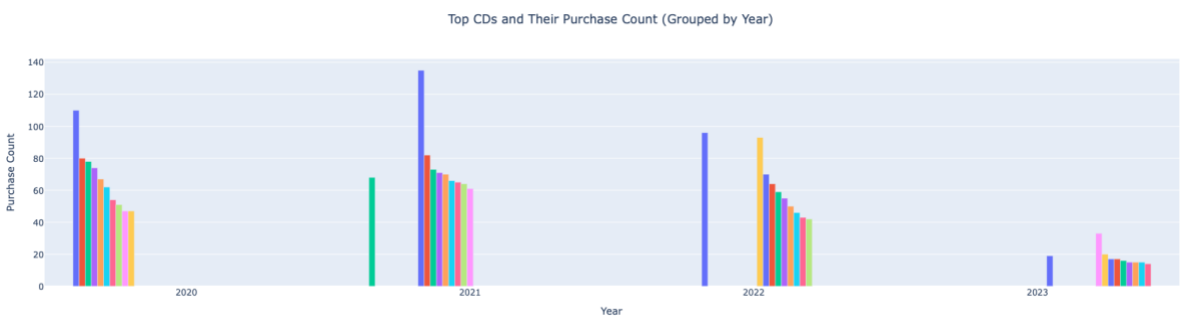
These findings lend support to the hypothesis that higher-priced albums aren't just purchased more they also tend to attract more favorable user ratings, reinforcing the relationship between perceived quality and both economic and emotional investment in digital music.

Plot 7 – (see Interactive plot [here](#))



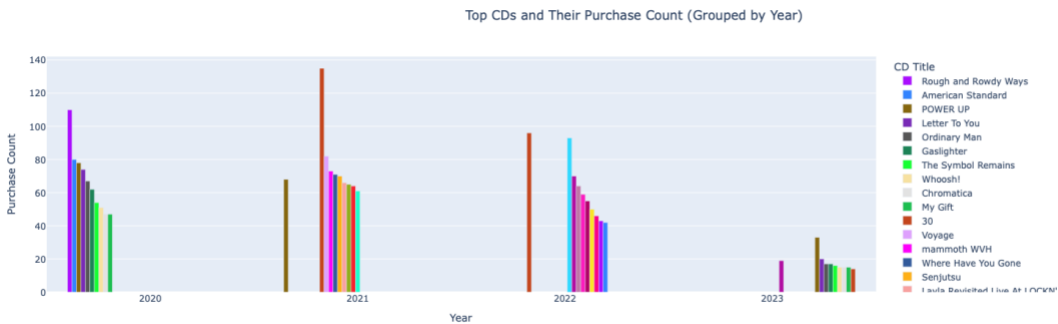
However, we wanted to further explore in more clear form if some CD repeats over the year,

Plot 7.1 – (see interactive plot [here](#))



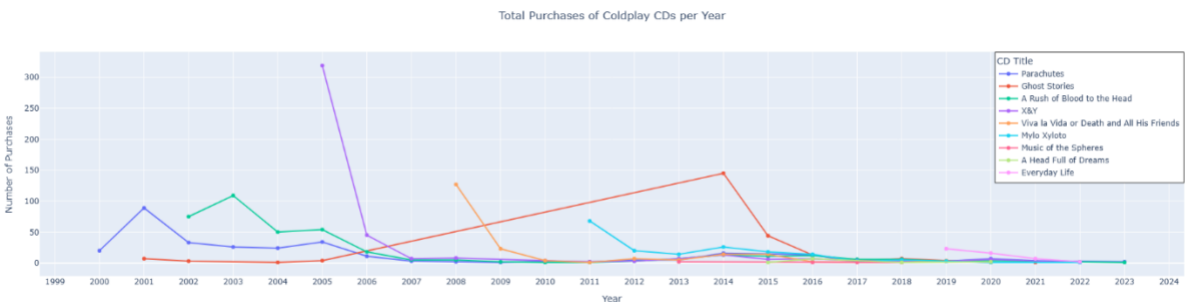
Plot 7.2 – (see interactive plot [here](#))

with legend to see the CD names



Next, we wanted to explore purchase behaviour of Coldplay band, we created a list of their CD's title and release year, then showed on a graph the purchases over the years of those CD's.

Plot 8 – (see interactive plot [here](#))

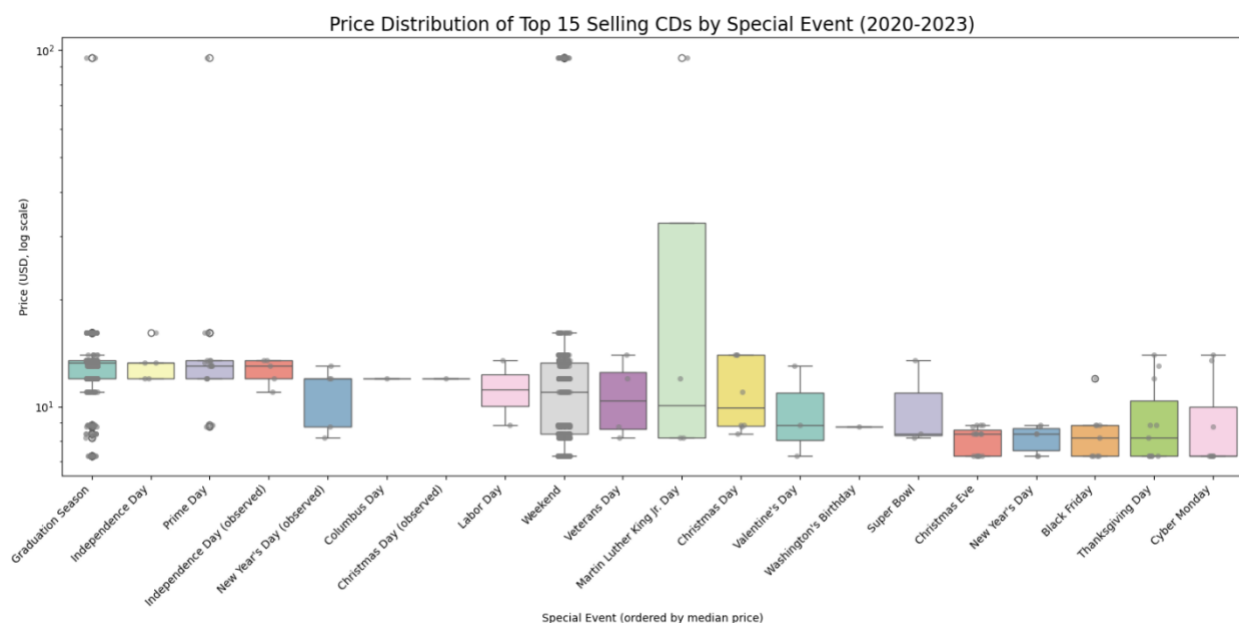


We found that for Coldplay album “X&Y” the year it was released (2005) as well was the year with the highest number of purchases, same for “Viva la Vida” , however we spotted that “Ghost Stories” that was released in 2014, had value of purchases from 2001.

Hypothesis 4 - price distribution by special events

How events over time like holidays etc effect price distribution in top 15 selling CDs?
for this we filtered the data to focus on recent period 2020-2023 and only the top 15 CDs
then drop rows where 'special_event' is None as they don't represent a specific event.

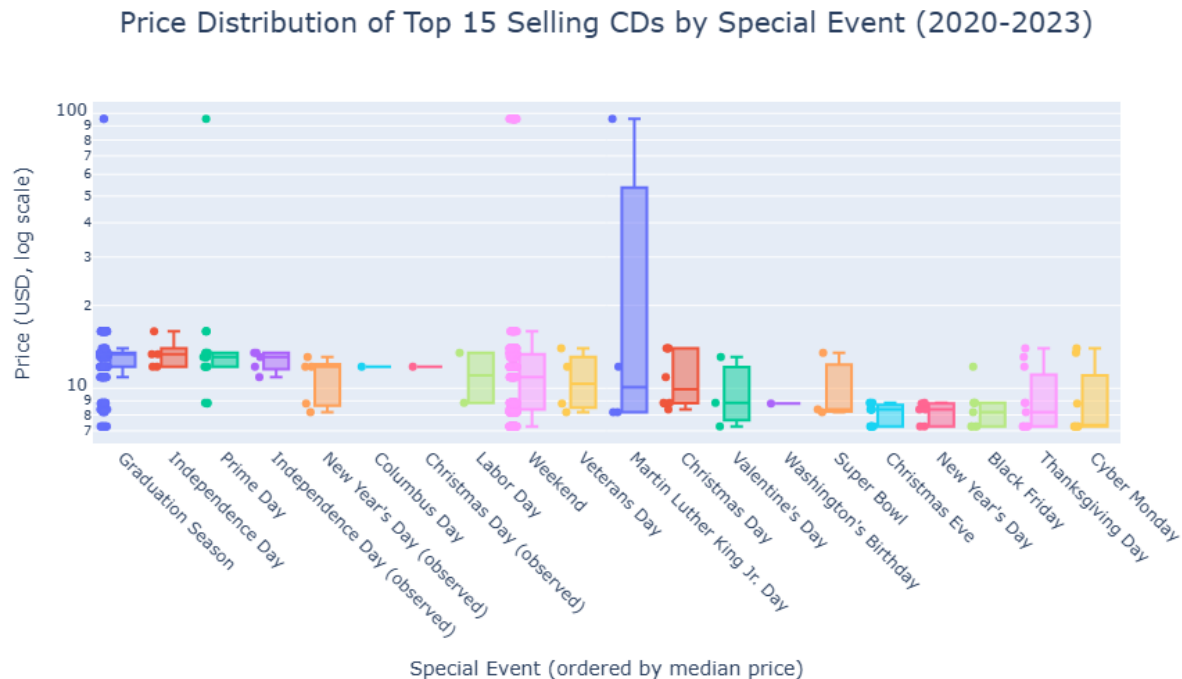
Plot 9 - Raw Price Distributions by Event



Across the calendar, special events exert a powerful influence on the pricing of the top 15 best-selling CDs: the deepest, most uniform discounts appear on true bargain days like Black Friday and Cyber Monday, where median prices plunge to around \$8–\$10 and virtually eliminate high-end outliers. By contrast, traditional gift-giving holidays such as Christmas and Valentine’s Day maintain moderate medians (roughly \$12–\$13) but allow for a broader range of prices as retailers balance full-price new releases with spot promotions on back-catalog titles.

Prime Day and Graduation Season fall somewhere in between, offering both significant markdowns on some blockbusters and near–full-price listings on others, while non-sale public holidays like Independence Day or Veterans Day often serve as launch windows for deluxe or collectible editions creating dramatic price spikes up to \$80–\$90 alongside standard releases. In this way, just as we observed in user ratings, CD prices “flow” with the calendar: deeply discounted during mass-market sale events, stratified during gift-driven seasons, and spiking on public holidays when premium editions debut.

Plot 9.1 – (see interactive plot [here](#))



When we re-plot CD prices on a logarithmic scale and order events by their median, a clear pattern emerges: non-sale public holidays such as Independence Day (observed) and Veterans Day stand out for their high-end spikes those \$80–\$90 dots represent premium vinyl and deluxe box-set releases timed to those occasions—whereas true “deal days” like Black Friday and Cyber Monday collapse prices to their lowest medians and nearly erase any expensive outliers, reflecting the broadest, deepest markdowns. Holidays such as Christmas and Valentine’s Day occupy the middle ground, with moderate median prices punctuated by a few outliers as retailers balance selective discounts on back catalogue titles against maintaining a slight premium on new holiday releases. Prime Day and similar crossover events fall between these extremes: their medians skew left, indicating meaningful bargains, yet they still accommodate occasional higher-priced items, underscoring their hybrid sale-and-gift character.

Special-event timing profoundly shapes the price landscape for top-selling CDs:

- **Deep-discount events** compress both average and high-end prices, driving volume by making blockbuster titles truly bargain-priced.
- **Traditional gift holidays** allow for more **price stratification**: retailers balance promotional discounts on older or less popular titles with **full price “must-have” new releases**.
- **Non-sale public holidays** often serve as launch windows for **deluxe, collectible editions**, creating wide price spreads and occasional spikes far above the typical \$10–\$15 range.

In short, just as we saw with ratings (H1–H2), price isn’t static: it flows with the calendar. Customers pay premium for special editions on standard holidays, whereas they clutch bargains for mass-market hits on sale days. This interplay of **event timing**, **product positioning**, and **consumer expectation** completes our picture of how both **sentiment** and **spending** dance to the beat of the calendar.

Conclusions

Understanding user ratings is essential for e-commerce platforms like Amazon and for artists, as these ratings reflect customer satisfaction and product success. In analyzing Amazon's Digital Music Category, we explored user ratings systematically, seeking insights into distribution patterns, temporal trends, price relationships, and rating consistency across popular titles.

Our analysis confirmed several key hypotheses and offered nuanced insights:

Firstly, ratings showed a clear positive bias, overwhelmingly clustering around 4 and 5 stars.

About 90% of ratings were positive, confirming that digital music purchasers either genuinely experience high satisfaction or exhibit strong leniency, while dissatisfied customers rarely leave reviews. Furthermore, this positivity bias is notably concentrated among a few blockbuster CDs that capture the lion's share of attention and acclaim.

Secondly, we observed significant temporal trends in average user ratings over the years.

Starting from high ratings in the early days of Amazon's music review system (late 1990s), ratings declined substantially in the early to mid-2000s, likely reflecting growing consumer expectations and critical mass as the platform expanded. However, from the late 2000s onward, ratings steadily recovered, stabilizing at high levels since the mid-2010s, likely driven by improvements in Amazon's platform, review policies, and refined music curation.

Our examination of pricing reinforced the hypothesis that higher-priced digital music products tend to attract higher average ratings, as premium albums consistently occupied leading positions in both ratings and sales. Special-event analyses revealed that pricing fluctuates significantly depending on calendar events. Major discount days such as Black Friday and Cyber Monday dramatically reduced prices and limited price variation, while holidays and public events led to increased pricing variance due to the presence of premium editions.

Amazon's digital music ratings ecosystem is highly positive, deeply concentrated around top-selling CDs, influenced by price and special events, and exhibits distinct historical patterns shaped by consumer behavior and platform evolution. These insights provide valuable strategic directions for artists, marketers, and e-commerce managers seeking to optimize product positioning, timing, and pricing strategies.

Understanding user ratings is essential for e-commerce platforms like Amazon and for artists, as these ratings reflect customer satisfaction and product success. In analyzing Amazon's Digital Music Category, we explored user ratings systematically, seeking insights into distribution patterns, temporal trends, price relationships, and rating consistency across popular titles.

Our analysis confirmed several key hypotheses and offered nuanced insights:

H1: Ratings show a clear positive bias, overwhelmingly clustering around 4 and 5 stars. About 90% of ratings are positive, confirming that digital music purchasers either genuinely experience high satisfaction or exhibit strong leniency, while dissatisfied customers rarely leave reviews. Furthermore, this positivity bias is notably concentrated among a few blockbuster CDs that capture the lion's share of attention and acclaim.

H2: There are significant temporal trends in average user ratings over the years. Starting from high ratings in the early days of Amazon's music review system (late 1990s), ratings declined substantially in the early to mid-2000s, likely reflecting growing consumer expectations and critical mass as the platform expanded. However, from the late 2000s onward, ratings steadily recovered, stabilizing at high levels since the mid-2010s, likely driven by improvements in Amazon's platform, review policies, and refined music curation.

H3: Higher-priced digital music products tend to attract higher average ratings, as premium albums consistently occupy leading positions in both ratings and sales.

H4: The distribution of CD prices is significantly shaped by the timing of special events. Deep-discount days such as Black Friday and Cyber Monday consistently yield the lowest median prices (around \$8–\$10) with minimal price variation, reflecting platform-wide markdown strategies. In contrast, gift-driven holidays such as Christmas and Valentine's Day maintain moderate median prices (\$12–\$13) with wider variation, as retailers blend selective discounts with full-price premiums on in-demand releases. Prime Day and Graduation Season strike a middle balance, offering mixed pricing strategies, while non-sale public holidays such as Independence Day and Veterans Day often trigger the release of deluxe or collectible editions, producing high-end spikes up to \$80–\$90. This cyclical price behavior demonstrates that, much like rating patterns, CD pricing “flows” with the calendar—discounted heavily during mass-market sales, stratified during gift seasons, and elevated during special-edition launches.

In summary, Amazon's digital music ratings ecosystem is highly positive, deeply concentrated around top-selling CDs, influenced by price and special events, and exhibits distinct historical patterns shaped by consumer behavior and platform evolution. These insights provide valuable strategic directions for artists, marketers, and e-commerce managers seeking to optimize product positioning, timing, and pricing strategies.